

Practical 6: Stationarity Testing and Transformation

Question: Select any time series (e.g., stock prices, rainfall). Perform a stationarity check using the Augmented Dickey-Fuller (ADF) test. Plot the rolling mean and rolling standard deviation. Convert the series into a stationary series using differencing and show the before/after results.

6.1 Objective

- To check if a time series is stationary using Rolling Statistics and the Augmented Dickey-Fuller (ADF) test.
- To visualize the rolling mean and standard deviation.
- To transform a non-stationary series into a stationary one using differencing.

6.2 Theory

- **Stationarity:** A series is stationary if its statistical properties (mean, variance) remain constant over time. Most forecasting models (like ARIMA) assume stationarity.
- **ADF Test:** A statistical test for stationarity.
 - **Null Hypothesis (H0):** The series is non-stationary (has a unit root).
 - **Alternative Hypothesis (H1):** The series is stationary.
 - **Interpretation:** If p-value < 0.05, reject H0 (Series is Stationary).
- **Differencing:** Computing the difference between consecutive observations ($Y_t - Y_{t-1}$) to remove trends and stabilize the mean.

6.3 Code Implementation

```
# 1. Function to Check Stationarity
def check_stationarity(timeseries):
    # Calculate Rolling Statistics
    rolmean = timeseries.rolling(window=12).mean()
    rolstd = timeseries.rolling(window=12).std()

    # Plot Rolling Statistics
    plt.figure(figsize=(12, 5))
    plt.plot(timeseries, label='Original')
    plt.plot(rolmean, color='red', label='Rolling Mean')
    plt.plot(rolstd, color='black', label='Rolling Std Dev')
    plt.legend()
    plt.show()

    # Perform ADF Test
    result = adfuller(timeseries)
    print(f'ADF Statistic: {result[0]}')
    print(f'p-value: {result[1]}')
    if result[1] <= 0.05:
        print("Conclusion: Series is Stationary")
    else:
        print("Conclusion: Series is Non-Stationary")

# 2. Check Original Series
# Assuming 'ts_data' is your loaded time series
```

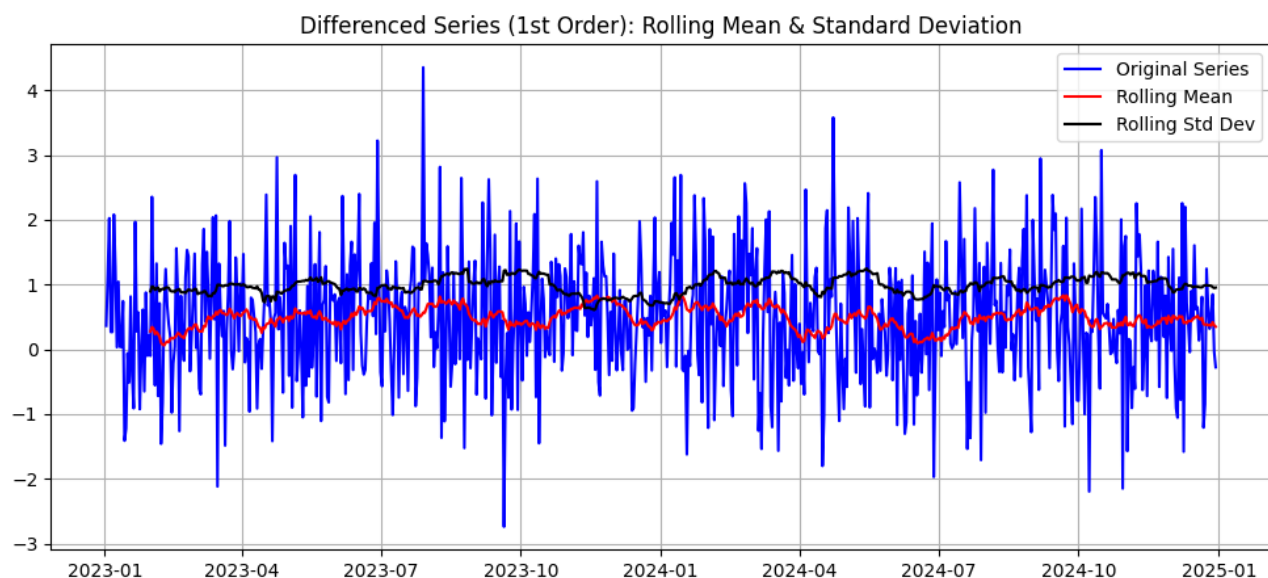
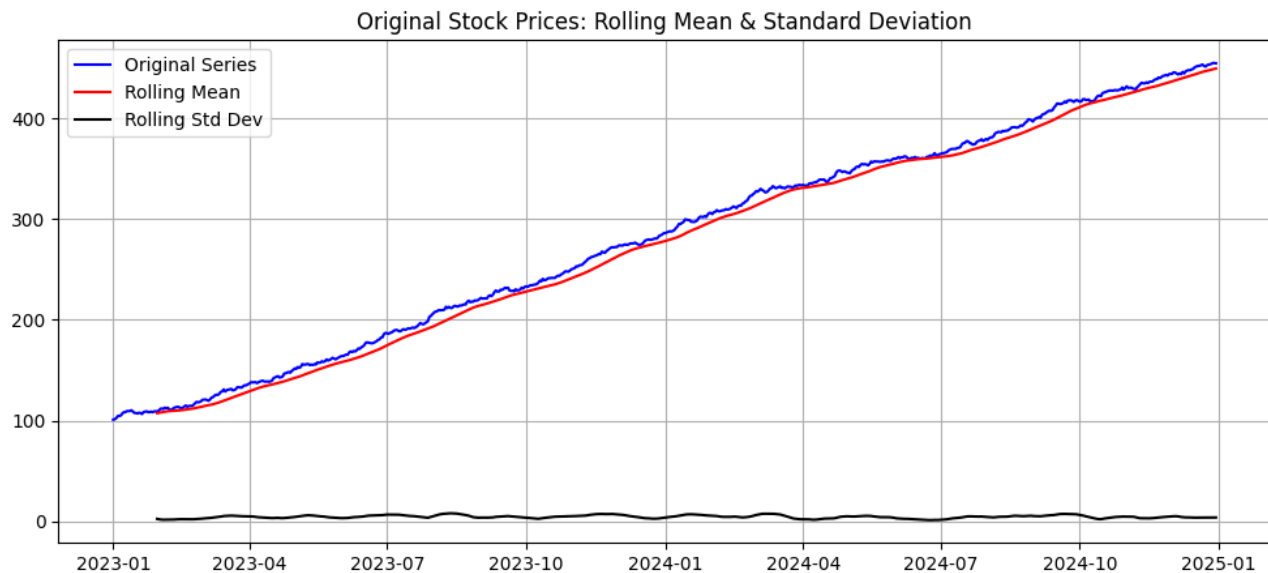
```
check_stationarity(ts_data)
```

```
# 3. Apply Differencing
```

```
df_diff = ts_data.diff().dropna()
```

```
# 4. Check Differenced Series
```

```
check_stationarity(df_diff)
```



6.4 Interpretation

- **Before Differencing:** The rolling mean typically shows a trend (moving up or down), and the ADF test yields a p-value > 0.05 , indicating the series is **non-stationary**.
- **After Differencing:** The rolling mean becomes roughly constant (flat), and the ADF test yields a p-value < 0.05 , indicating the series has successfully become **stationary**.